



Mecha-Basketball 3D

a dumb game about robots playing basketball

*Interactive Graphics 2025/26, Course by Professor Marco Schaerf
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First Look at the Game: 3 Robot Classes to play





How to play

Input	Effect
Mouse/Cursor	Aiming
WASD	Movement
Left Shift	Special Move
Left Mouse	Shoot!
Right Mouse	Handle the ball and Aim
Q	BLOCK
E	STEAL





Development Environment, Libraries and Assets

Stack:

- HTML, CSS
- Javascript
- **Three.js**
- **Vite**

Rendering Options

- Shadow Map : **PCFSoftShadowMap**
- toneMapping: **ACESFilmicToneMapping**
- outputColorSpace: **SRGBColorSpace**

External Free Assets (From Sketchfab)

- [Basketball Court and Textures](#) from [Stew004](#)
- [Basketball ball and Textures](#) from [ikagogava](#)



Models & Hierarchical Structure (1/2) [Manipulators]

Legend	MESH	GROUP	OBJECT-3D
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```
root
• wheelsGroup
  ○ wheel [1...4]
• disc
• base
  ○ basejoint
  ○ link1Group
    ■ link1
    ■ elbow
      ● elbowJoint
      ● link2
  • wrist
    ○ endEffector
    ○ paddleGroup
      ■ paddle (x2)
      ■ paddleCenter
      ■ ballRestPoint
```



```
root
• legsGroup
  --- for each leg ---
  • hipYaw
    ○ hip
      ■ hipJoint
      ■ thigh
      ■ knee
      ● kneeJoint
      ● shin
      • ankle
      ○ foot
  -----
  ... then same structure
  for the ARM
```



Models & Hierarchical Structure (2/2)

[Drone]

Legend	MESH	GROUP	OBJECT-3D
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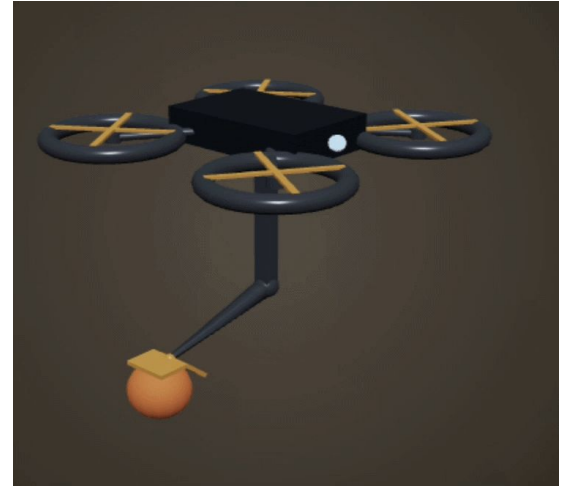
```
root
● bodyGroup
  ○ hull
    ● noseLight
    ● arm [1...4]
      ● pivot [1...4]
        ○ guard
        ○ hub
        ○ bladeA
        ○ bladeB
      ● base
        ○ armFlip
          ■ baseJoint...
        ... then same structure
        for the rest of the ARM
```





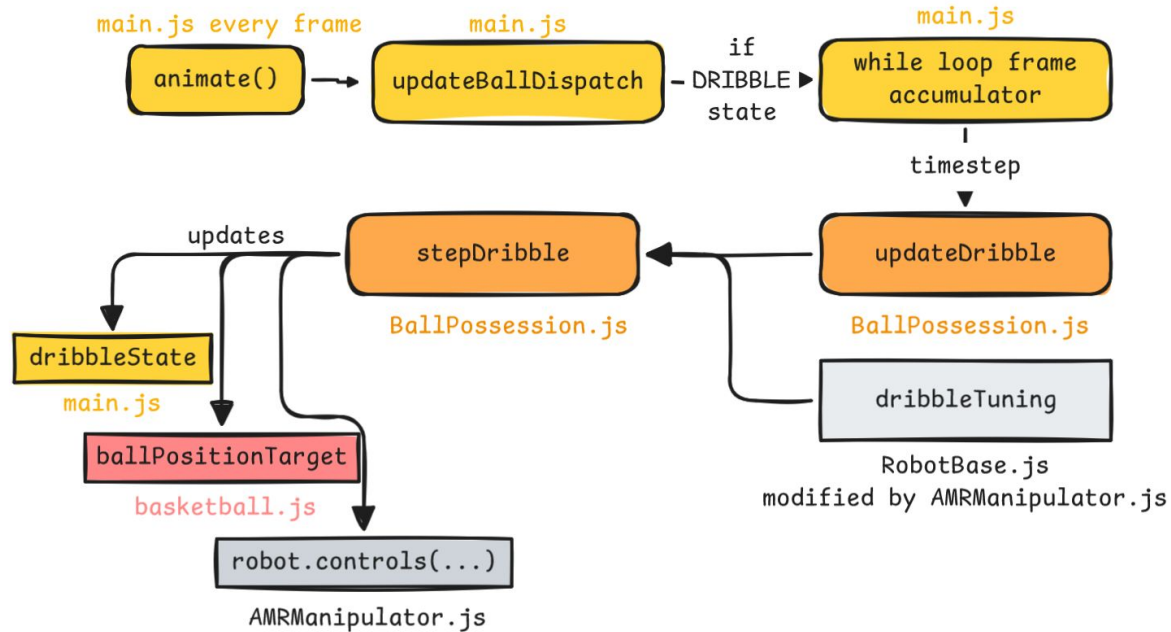
Animations

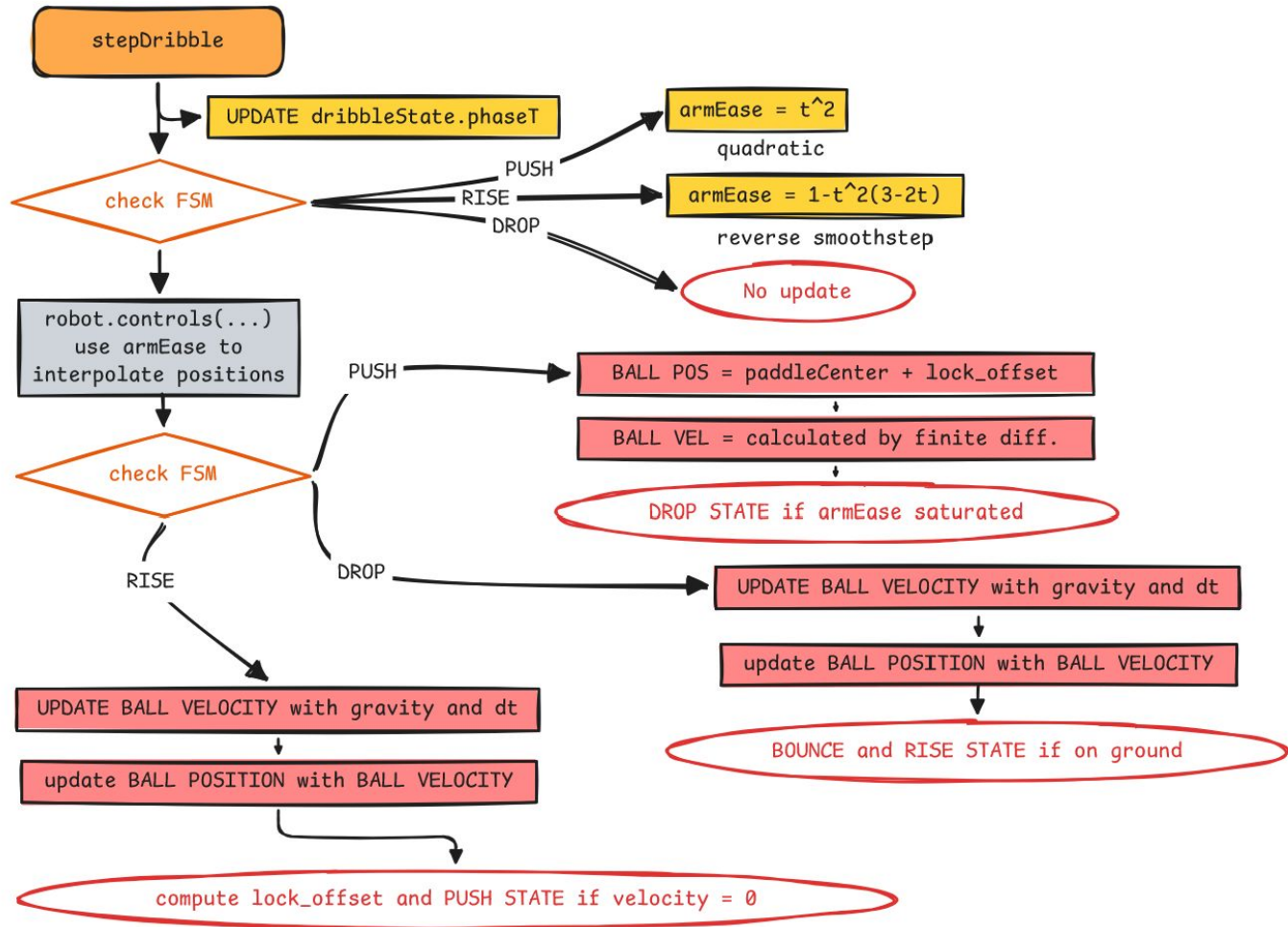
[Example: Dribbling Animation]



Animations

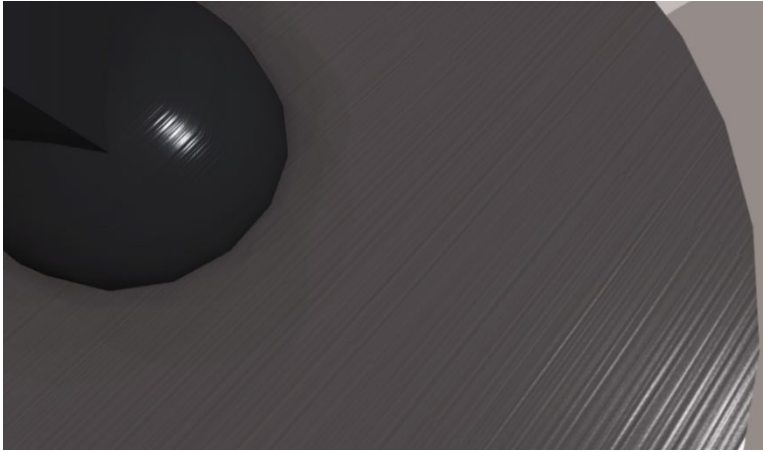
[Example: Dribble for AMR Manipulator]





Materials & Textures (1/2)

Two kinds: **downloaded** (*ball color map, court diffuse map*) and **procedurally generated**.



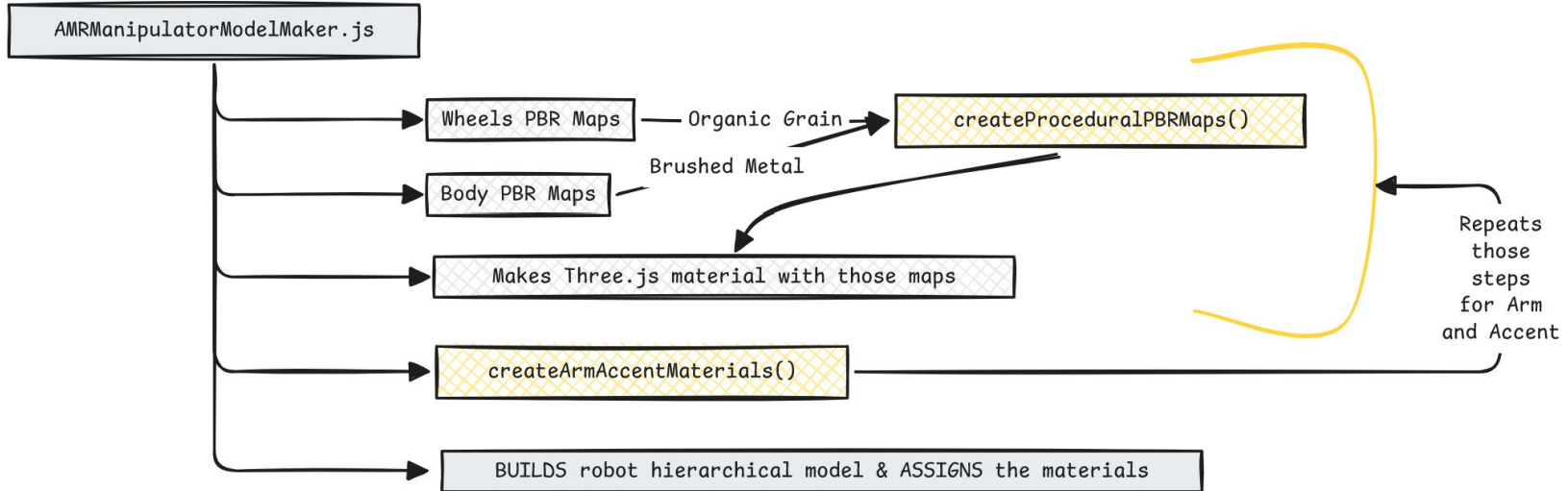
Generated PBR Texture for the Robot Body



Imported Texture for the Basketball



Materials & Textures (2/2)



Lights

1 **HemisphereLight**, 1 **DirectionalLight** (4096^2 shadow map, frustum sized to cover the court), 4 **PointLight** (lamp posts) and 2 **SpotLight** (hoop fixtures, lit only at Sunset/Night). Colors, intensities and sun position follow a TimeOfDay **preset table** (*Sunrise/Day/Sunset/Night*).



Shooting System (1/2)

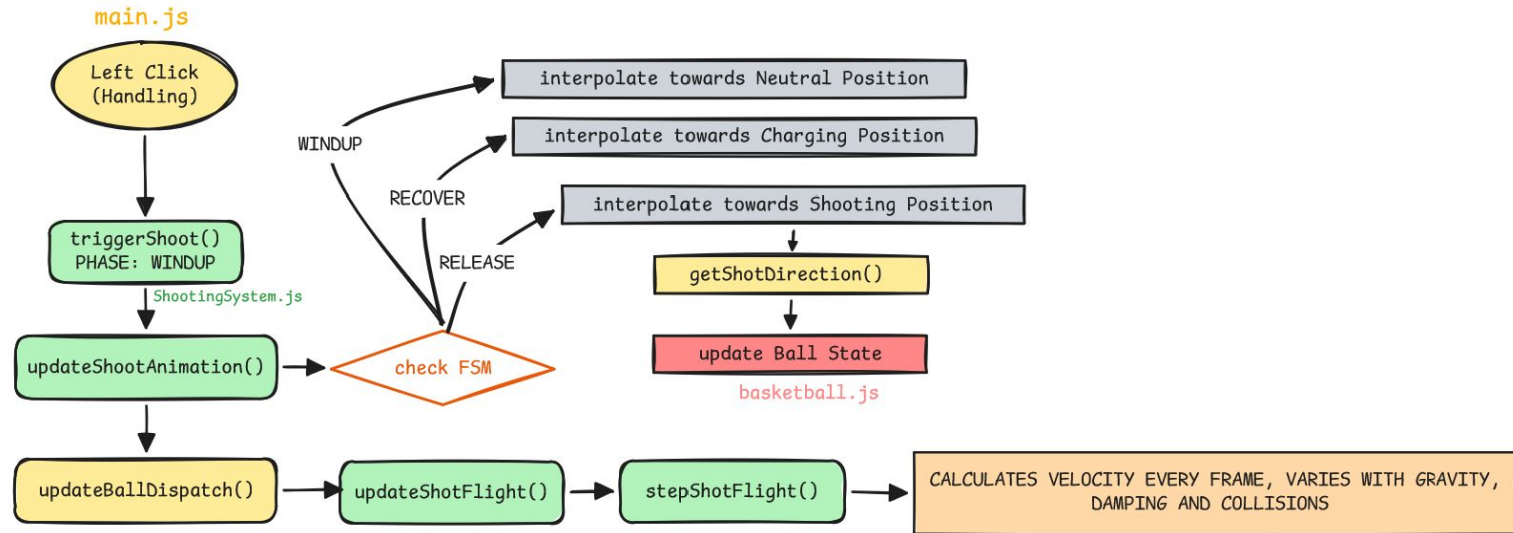


The Shooting Systems simulates every frame an entire trajectory (shown by a 3D Tube Mesh) that is supposed to help aiming at the hoop.

We have two tubes, one is gray, the other is **GREEN** if it intersects with the hoop, **BLUE** otherwise.



Shooting System (2/2)





Thank you for the attention!

[Play The Demo Here](#)

